T1. $xy' = -2\sqrt{xy} + y$

$y' = -2\sqrt{\frac{y}{x}} + \frac{y}{x}$

Let $v = \frac{y}{x}$, $y = vx$

$\cancel{v} + v'x = -2\sqrt{v} + \cancel{v}$

$\frac{dv}{dx} \cdot x = -2\sqrt{v}$

$\frac{1}{-2\sqrt{v}}\, dv = \frac{1}{x}\, dx$

$-\frac{1}{2}\int v^{-\frac{1}{2}}\, dv = \ln|x| + c$

$-\frac{1}{2} \cdot 2 \cdot v^{\frac{1}{2}} = \ln|x| + c$

$-v = (\ln|x| + c)^2$

$y = -x(\ln|x| + c)^2$ ✓

T2. $\frac{dy}{dx} = \frac{1}{xy(1 + x^2y^2)}$

$\frac{dx}{dy} = xy + x^3y^3$

$\frac{dx}{dy} = xy + x^3y^3$

Suppose $z = x^n$ $x = z^{\frac{1}{n}}$

$\frac{dz}{dy} = n \cdot x^{n-1} \cdot \frac{dx}{dy}$ & $\frac{dx}{dy} = \frac{1}{n} \cdot x^{1-n} \frac{dz}{dy} = \frac{1}{n} \cdot z^{\frac{1-n}{n}} \frac{dz}{dy}$

$\frac{1}{n} \cdot z^{\frac{1-n}{n}} \cdot \frac{dz}{dy} = z^{\frac{1}{n}} \cdot y + z^{\frac{3}{n}} y^3$

$\frac{dz}{dy} = nzy + n \cdot z^{1+\frac{2}{n}} \cdot y^3$

Let $1 + \frac{2}{n} = 0 \Rightarrow n = -2$

$z' = -2zy - 2y^3$

$z' + 2yz = -2y^3$

$M = e^{\int 2y\, dy} = e^{y^2}$

$z = e^{-y^2} \int e^{y^2} \cdot (-2y^3)\, dy$

$= -e^{-y^2} \int y^2\, d(e^{y^2})$

$= -e^{-y^2} \cdot (y^2 e^{y^2} - \int e^{y^2}\, d(y^2))$

$= -e^{-y^2} \cdot (y^2 e^{y^2} - e^{y^2} + c)$

$x^{-2} = -y^2 + 1 + ce^{-y^2}$

$y = (-y^2 + 1 + ce^{-y^2})^{-\frac{1}{2}}$

T3. $\underbrace{(e^x + 3y^2)}_{M}\, dx + \underbrace{2xy}_{N}\, dy = 0$

Verify, $M_y = 6y \neq N_x = 2y$

s.t. $\left((e^x + 3y^2)M\right)_y = \left((2xy) \cdot M\right)_x$

$$\mu \cdot 6y + \mu_y \cdot (e^x + 3y^2) = 2y \cdot \mu + \mu_x \cdot (2xy)$$

$$4y\mu = 2xy \cdot \mu_x - (e^x + 3y^2)\mu_y$$

Suppose μ is function of x, $\mu_y = 0$.

$$\Rightarrow 2\mu = x\,\mu_x$$

$$\Rightarrow 2\mu = x \cdot \frac{\partial \mu}{\partial x} \Rightarrow \frac{1}{x}dx = \frac{1}{2\mu}d\mu \Rightarrow 2\ln|x| = \ln|\mu|$$

$$\mu = x^2$$

$$x^2(e^x + 3y^2)\,dx + 2x^3y\,dy = 0$$

$$\varphi(x,y) = \int x^2(e^x + 3y^2)\,dx + h(y) = x^2e^x - 2(e^x x - e^x) + x^3y^2 + h(y)$$

$$\frac{\partial \varphi}{\partial y} = 2yx^3 + h'(y) = 2x^3y \Rightarrow h'(y) = 0, \quad h(y) = c.$$

s.t. $\varphi(x,y) = x^2e^x - 2e^x x + 2e^x + x^3y^2 = C$

T4. $xy' = 1 + (y')^2$

$$(y')^2 - xy' + 1 = 0$$

$$y' = \frac{x \pm \sqrt{x^2 - 4|1|}}{2} = \frac{1}{2}\cdot(x \pm \sqrt{x^2-4})$$

$$y = \int \frac{1}{2}x \pm \frac{1}{2}\sqrt{x^2-4}\,dx = \frac{1}{4}x^2 \pm \frac{1}{2}\int\sqrt{x^2-4}\,dx,$$

$$= \frac{1}{4}x^2 \pm \frac{1}{2}\cdot\left(x\cdot\sqrt{\frac{x^2-4}{4}} \mp 2\cosh^{-1}\left(\frac{x}{2}\right)\right)$$

$$= \frac{1}{4}x^2 \pm \frac{1}{4}x\sqrt{x^2-4} \mp \cosh^{-1}\left(\frac{x}{2}\right) + c$$

$$x^2 - 4\sinh^2(t) = 4$$

$$\cosh^2 x - \sinh^2 x = 4$$

$$x = 2\cosh t,\ dx = 2\sinh t\,dt$$

$$\int\sqrt{x^2-4}\,dx = \int 2\sinh(t)\cdot 2\sinh(t)\,dt = 4\int \sinh^2(t)\,dt$$

$$= 4\cdot\frac{1}{2}\cdot(\sinh(t)\cosh(t) - t) = 2\sinh(t)\cosh(t) - 2t$$

$$= x\cdot\sqrt{\frac{x^2-4}{4}} - 2\cosh^{-1}\left(\frac{x}{2}\right)$$

$$\int \sinh(t)\,d(\cosh(t)) = \sinh t\cosh t - \int \cosh(t)\,d(\sinh(t)) = \sinh(t)\cosh(t) - \int \cosh^2(t)\,dt$$

$$= \sinh(t)\cosh(t) - C\int 1 + \sinh^2(t)\,dt$$

$$\int \sinh^2 = \sinh\cosh - t - \int \sinh^2 dt$$

$$\Rightarrow \int \sinh^2(t) = \frac{1}{2}(\sinh(t)\cosh(t) - t)$$

$$\cosh^2 - \sinh^2 = 1 + \sinh^2$$

T5. $\dfrac{dy}{dx} + \dfrac{1 + xy^3}{1 + x^3y} = 0 \qquad \Rightarrow \dfrac{dy}{dx} = -\dfrac{1+xy^3}{1+x^3y}$

$$\begin{cases} v = xy & dv = dx + dy \\ u = x + y & du = y\,dx + x\,dy \end{cases}$$

$$\frac{du}{dv} = \frac{dx + dy}{y\,dx + x\,dy} = \frac{1 + \frac{dy}{dx}}{y + x\frac{dy}{dx}} = \frac{1 - \frac{1+xy^3}{1+x^3y}}{y - x\cdot\frac{1+xy^3}{1+x^3y}} = \frac{1 + x^3y - 1 - xy^3}{y + x^3y^2 - x - x^3y^3} = \frac{x^3y - xy^3}{y - x + x^3y^2(x-y)} = \frac{xy(x-y)(x+y)}{(x-y)(x^3y^2-1)} = \frac{xy\cancel{(x-y)}(x+y)}{\cancel{(x-y)}(xy-1)(xy+1)} = \frac{uv}{v^2-1}$$

$$\int \frac{1}{u}\,du = \frac{1}{2}\int\frac{2v}{v^2-1}\,dv$$

$$2\ln|u| = \ln|v^2-1| + c$$

$$\ln\frac{u^2}{|v^2-1|} = c$$

$$\frac{u^2}{|v^2-1|} = e^c = C$$

$$(x+y)^2 = C|x^2y^2-1|$$

$$C(x+y) = \sqrt{x^2y^2-1}, \quad C \in \mathbb{R}$$

$\boxed{\text{Euler equation}}$

$$x^n y^{(n)} + P_1 x^{n-1} y^{(n-1)} + \cdots + P_{n-1} x y' + P_n y = f(x)$$

Let $x = e^t$, $\dfrac{dy}{dx} = \dfrac{dy}{dt}\dfrac{dt}{dx} = \dfrac{1}{x}\cdot\dfrac{dy}{dt}$

Notice that $\frac{dt}{dx} = \frac{1}{x}$

$$\frac{d^2y}{dx^2} = \frac{d\frac{dy}{dx}}{dx} = \frac{d\left(\frac{1}{x}\cdot\frac{dy}{dt}\right)}{dx} = \frac{-\frac{1}{x^2}\cdot dx\cdot\frac{dy}{dt} + d\left(\frac{dy}{dt}\right)\cdot\frac{1}{x}}{dx} = -\frac{1}{x^2}\frac{dy}{dt} + \frac{d\left(\frac{dy}{dt}\right)}{dt}\cdot\frac{dt}{dx}\cdot\frac{1}{x} = \frac{1}{x^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$$

Suppose $D = \frac{d}{dt}$

$$x^2y'' = D(D-1)y$$

$$xy' = Dy$$

$$\frac{d^3y}{dx^3} = \frac{d\left(\frac{d^2y}{dx^2}\right)}{dx} = \frac{d\left(\frac{1}{x^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)\right)}{dx} = \frac{-2\frac{1}{x^3}dx\cdot\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right) + \frac{1}{x^2}\cdot d\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)}{dx} = -\frac{2}{x^3}\cdot\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right) + \frac{1}{x^2}\cdot\frac{d\left(\frac{d^2y}{dt^2}\right)}{dt}\cdot\frac{dt}{dx} - \frac{1}{x^2}\cdot\frac{d\left(\frac{dy}{dt}\right)}{dt}\cdot\frac{dt}{dx} = -\frac{2}{x^3}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right) + \frac{1}{x^3}\frac{d^3y}{dt^3} - \frac{1}{x^3}\frac{d^2y}{dt^2}$$

$$= \frac{1}{x^3}\left(-2\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + \frac{d^3y}{dt^3} - \frac{d^2y}{dt^2}\right)$$

$$= \frac{1}{x^3}\left(\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 2\frac{dy}{dt}\right)$$

$$\Rightarrow x^3 y''' = D(D-1)(D-2)y$$

$$\vdots$$

$$x^k y^{(k)} = D\cdot(D-1)\cdots(D-k+1)y$$

T6. $t^2x'' - tx' + 2x = t\ln t$.

$t = e^y \qquad D = \frac{d}{dy}$

$tx' = Dx$

$t^2x'' = D(D-1)x$

$$\left(\frac{d^2x}{dy^2} - \frac{dx}{dy}\right) - \frac{dx}{dy} + 2x = e^y\cdot y$$

$$x'' - 2x' + 2x = e^y\cdot y$$

$$r_1 = \frac{2\pm\sqrt{4-4\times 2}}{2} = \frac{2\pm 2i}{2} = 1\pm i$$

Let $x = c\cdot e^y\cdot y$, $x' = c\cdot(1+y)e^y$, $x'' = c(2+y)e^y$

$$c\cdot(2+y)e^y - 2c\cdot(1+y)e^y + 2cye^y = e^y\cdot y$$

$$c(2+y) - 2c(1+y) + 2cy = y$$

$$2c + cy - \cancel{2c} - \cancel{2cy} + \cancel{2cy} = y$$

$$c = 1$$

$\Rightarrow x = e^y\cdot y + c_1e^y\cos y + c_2e^y\sin y$, $c_1, c_2 \in \mathbb{R}$.

$x = t\cdot\ln t + c_1t\cos\ln t + c_2t\sin\ln t$,

$x''' - 3x'' + 3x' - x = t - 3$.

$x = at^3 + bt^2 + ct + d$

$x' = 3at^2 + 2bt + c$

$x'' = 6at + 2b$

$x''' = 6a$

$$6a - 3\cdot(6at+2b) + 3\cdot(3at^2+2bt+c) - (at^3+bt^2+ct+d) = t-3$$

$$\cancel{6a} - \cancel{18at} - \cancel{6b} + \cancel{9at^2} + \cancel{6bt} + \cancel{3c} - \cancel{at^3} - \cancel{bt^2} - \cancel{ct} - \cancel{d} = t-3$$

$$-at^3 + (9a-b)t^2 + (6b-18a-c)t + 6a - 6b + 3c - d = t-3$$

$$\begin{cases} -a = 0 \\ 9a - b = 0 \\ 6b - 18a - c = 1 \\ 6a - 6b + 3c - d = -3 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = 0 \\ c = -1 \\ d = 0 \end{cases}$$

s.t $x = -t$.

$r^3 - 3r^2 + 3r - 1 = 0$

$(r-1)^3 = 0$

$r = 1, 1, 1$

$x = (c_1t^2 + c_2t + c_3)e^t - t$, $c_1, c_2, c_3 \in \mathbb{R}$

2-i

T8. $x' = \begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix} x$

$$\det(A-\lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) + 2 = 0 \Rightarrow \lambda^2 - 4\lambda + 5 = 0, \quad \lambda = \frac{4\pm 2i}{2} = 2\pm i$$

$2+i$

$$\begin{vmatrix} 1-2-i & 1 \\ -2 & 3-2-i \end{vmatrix} = \begin{vmatrix} -1-i & 1 \\ -2 & 1-i \end{vmatrix} \Rightarrow \begin{pmatrix} 1 \\ 1+i \end{pmatrix}$$

$$x_c(t) = e^{(2+i)t}\begin{pmatrix} 1 \\ 1+i \end{pmatrix} = e^{2t}\cdot(\cos t + i\sin t)\cdot\begin{pmatrix} 1 \\ 1+i \end{pmatrix} = e^{2t}\begin{pmatrix} \cos t + i\sin t \\ \cos t - \sin t + i\cos t + i\sin t \end{pmatrix}$$

$$= e^{2t}\begin{pmatrix} \cos t \\ \cos t - \sin t \end{pmatrix} + ie^{2t}\begin{pmatrix} \sin t \\ \cos t + \sin t \end{pmatrix}$$

$$\Rightarrow x' = \begin{pmatrix} e^{2t}\cos t & e^{2t}\sin t \\ e^{2t}\cos t - e^{2t}\sin t & e^{2t}\cos t + e^{2t}\sin t \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

T 9. $x' = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} x + \begin{pmatrix} e^{2t} \\ 0 \end{pmatrix}$

$x_2' - 2x_2$

$x_1' = 2x_1 + x_2 + e^{2t} \Rightarrow x_2'' = 2x_2' + x_1' = 2x_2' + x_2 + 2x_1 + e^{2t}$

$x_2' = x_1 + 2x_2$

$x_2'' - 4x_2' + 3x_2 = e^{2t}$

$(r-3)(r-1) = 0$

$r = 3, 1$

$x_2 = Ae^{2t}$

$x_2' = 2Ae^{2t}$

$x_2'' = 4Ae^{2t}$

$4Ae^{2t} - 8Ae^{2t} + 3Ae^{2t} = e^{2t}$

$-Ae^{2t} = e^{2t}$

$\boxed{A = -1}$

$x_2 = c_1 e^{3t} + c_2 e^{t} - e^{2t}$

$x_1 = x_2' - 2x_2 = (3c_1 e^{3t} + c_2 e^{t} - 2e^{2t}) - 2(c_1 e^{3t} + c_2 e^{t} - e^{2t}) = c_1 e^{3t} - c_2 e^{t}$

$$x = \begin{pmatrix} c_1 e^{3t} - c_2 e^{t} \\ c_1 e^{3t} + c_2 e^{t} - e^{2t} \end{pmatrix} = \begin{pmatrix} e^{3t} & -e^{t} \\ e^{3t} & e^{t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ -e^{2t} \end{pmatrix}$$